

# Decision Memo: A JEPA-Style World Model for the Digital Liver

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## 1. Where I landed

I built a JEPA-style world model alongside four peer models, all on one shared decoder and one by-construction constraint head, so the comparison isolates what the learned representation contributes, not the constraint machinery. Three findings organise the memo.

**First, the hard constraints are solved by construction, not by the loss.** A decoder head that adds a strictly non-negative increment to each one-directional field makes monotonicity and every bound a property of the parameterisation, not something the training objective has to discover. The violation rate over 7,200 held-out transitions is exactly zero, and an inverse round-trip test shows the head can still represent any valid transition to within  $10^{-3}$ , so the guarantee costs no reachable accuracy. The malignancy coupling, where the hazard accumulates only from sustained fibrosis and cholestasis, is enforced the same way, so no learned accumulation rate is left free to drift from the truth.

**Second, on this clean, fully observed generator the raw state  $x(t)$  is a near-sufficient statistic, so a learned latent buys almost no accuracy, and I can show why.** All four peer models sit near the aleatoric noise floor (Table 1), and their confidence bands overlap, so the tie is real. Handing a model the true hidden susceptibility, which the oracle receives, doesn't help: it is no better than the memoryless baseline. Raw  $x(t)$  already decodes that susceptibility linearly at  $R^2 = 0.57$ , and the history and JEPA latents decode it no better, because the ratchet fields are essentially a running integral of it. A predictive latent has little to abstract that the state doesn't already carry.

**Third, and most important, the JEPA objective itself does not beat a plain supervised history encoder on any axis I can measure reliably here.** Supervised history matches or beats JEPA on accuracy, denoising, and decompensation detection. What genuinely helps is history itself: a window lets a model filter observation noise and read a progression rate from serial visits, and a memoryless model can do neither. An ablation that retrains the identical model with the latent-prediction term switched off isolates what the objective contributes: it raises the latent's effective rank by 1.3 (genuine anti-collapse) and improves the model by 0.009 against that ablation, but adds no gain in on-manifold behaviour. So the objective buys a collapse-resistant, auditable latent, not accuracy and not on-manifold-ness. I keep JEPA as the organising frame for those properties and say plainly that its accuracy case is unproven here.

| model                              | seed 0 | 3-seed mean $\pm$ sd                | note                                  |
|------------------------------------|--------|-------------------------------------|---------------------------------------|
| memoryless ( $x(t)$ is the latent) | 0.085  | $0.075 \pm 0.007$                   | the honest strong baseline            |
| supervised history                 | 0.078  | $0.072 \pm 0.005$                   | ties for best; JEPA does not beat it  |
| graph-attention encoder            | 0.077  | <b><math>0.070 \pm 0.005</math></b> | ties supervised; wins on auditability |
| JEPA                               | 0.087  | $0.082 \pm 0.005$                   | nominally the weakest predictor       |
| oracle (given true susceptibility) | 0.094  | $0.081 \pm 0.009$                   | no better; not a true upper bound     |
| aleatoric noise floor              | 0.081  | —                                   | irreducible flare noise               |

Table 1: In-distribution rollout mean absolute error on the held-out test split (months 12–48).

## 2. Why a predictive latent here, and what each peer establishes

I considered four alternatives to the JEPA route and built each one on the same head and decoder.

**The memoryless model treats  $x(t)$  itself as the latent.** It is the honest strong baseline, and it ties every other model in-distribution because the generator is very nearly Markov in  $x$ . Its structural ceiling, that it sees a single state, is where every later advantage of the history models comes from.

**A plain Neural-ODE, considered and rejected.** The dynamics aren't smooth: flares are discrete monthly jumps and an ERCP is a discrete step down, so a continuous vector field has to either smooth those events away or bolt on a separate discrete handler. One prior attempt indexed its ODE field with  $[t]$ , which is piecewise constant, so the continuous-time claim didn't survive the events.

**A supervised decoder over history is the peer that matters most.** It shares the JEPA encoder and differs only in dropping the latent-prediction objective, and across three seeds it is statistically tied for the best predictor I trained. That comparison states the third finding most cleanly: the JEPA objective adds a predictable, collapse-resistant latent, not a lower error.

**A causal graph-attention encoder is the recommended pairing, and I built it to measure rather than assert.** Each of the eight clinical fields is a node, and a single attention layer routes information across them, hard-masked to the disease's causal parents, which I read straight off the generator: the malignancy hazard attends to fibrosis and cholestasis, portal hypertension to fibrosis and inflammation, inflammatory activity and cholestasis to the flare, and the flare to strictures. Across three seeds it ties the recurrent encoder on accuracy,  $0.070 \pm 0.005$  against  $0.072 \pm 0.005$ , so it buys no accuracy. What it does buy is auditability: because its information flow is causal by construction, its attention weights are a readable account of the model's reasoning, and they recover the true drivers without being told to. So it earns its place at equal accuracy. I ship the simpler recurrent encoder as the default and keep the graph-attention encoder as the validated auditable alternative. An earlier draft claimed graph-attention couldn't earn its cost on eight nodes; building and measuring it proved that wrong, which

is exactly the standard the assignment sets against adopting architecture without an account of what it buys.

I commit to the JEPA frame, a history encoder that predicts a future latent which is then decoded by construction, with my eyes open. On this generator the latent-prediction objective isn't earning its complexity, and rather than pretend it already does, I name the regime where I would expect it to: noisy, irregularly sampled observation.

### 3. Keeping trajectories on the manifold, and what it costs

The decoder emits nine raw signals that a fixed head maps to a guaranteed-valid next state. Fibrosis, ductopenia, and portal hypertension are each the previous value plus a softplus of a raw signal, so they are monotone. Malignancy is the previous value plus a softplus increment multiplied by the current product of fibrosis and cholestasis. Strictures are the previous value plus a softplus increment minus an ERCP-gated relief term, so they can fall only in an ERCP month. Inflammatory activity, cholestasis, and the flare are bounded sigmoids. Cirrhosis is a fixed monotone function of fibrosis that is never stored, so it can never disagree with it.

**Engaging with the coupling, not only per-field monotonicity.** The assignment names three couplings, and they don't all admit a by-construction guarantee equally, so I treat them differently and say why. The malignancy hazard is a monotone accumulation of sustained  $F \cdot C$ , which reduces cleanly to a sign-constrained increment, so I put it directly in the head: malignancy's increment is a non-negative learned term times the current  $F \cdot C$ , so it can't rise unless both fibrosis and cholestasis are present, and can never fall. The other two are different in kind. A flare perturbs inflammatory activity and cholestasis together and then decays, and treatment suppresses both for responders; both interactions are soft, bidirectional, and transient, and neither reduces to a monotone increment. Forcing them into the head would either distort them or amount to re-deriving the generator's exact functional form, so I leave them to the learned decoder, which sees the flare and the treatment context, and I accept that their fidelity is checked empirically rather than guaranteed. That is the honest boundary of the approach: I close the one coupling that admits a clean structural form and say why the other two don't. The cost of the malignancy choice is that the model inherits that law rather than discovering it, so I don't count malignancy's behaviour as evidence that the model learned dynamics; the generalisation probes below test the parts that stay learned, the ratchet rates and the inferred susceptibility.

**The cost of the guarantee.** The head removes only the freedom to be invalid: the inverse round-trip test reconstructs any real transition to within  $10^{-3}$ , so no reachable accuracy is lost. The genuine cost is that the head guarantees validity, not fidelity; zero violations is not the same as staying on the manifold of realistic states. To probe that gap I train a manifold critic to separate real transitions from graded valid-but-wrong ones (successors from a different patient, from the wrong month of the same patient, and small off-manifold perturbations of the truth). It reaches an area under the curve of about 0.93 and scores every model's rollout below real transitions. The ranking among models rests on a single critic seed, so I treat it as suggestive; the firm signal is that an unconstrained baseline drifts off the manifold. The ablation also shows the latent-prediction objective doesn't improve this score, so I no longer claim on-manifold behaviour as something the objective buys.

### 4. Guarding the latent against collapse

The moment a learned latent enters, representation collapse becomes a live risk, so I designed against it rather than hope. Three mechanisms prevent it: the reconstruction loss, which makes full collapse unrepresentable because the latent must still decode to the state; a VICReg term that hinges each dimension's variance upward and penalises off-diagonal covariance, keeping the dimensions alive and decorrelated; and an exponential-moving-average, stop-gradient target encoder, which blocks the shortcut of encoding every input to one point. The metric that would catch a failure, independent of the loss, is the effective rank of the latent covariance: it is 4.77 of 16 on the reference seed, or  $4.2 \pm 0.4$  across seeds, against the data's measured intrinsic effective rank of 2.70. The latent uses more dimensions than the disease has, so it is neither collapsed nor padded. It dips toward 2 under early teacher forcing and recovers once VICReg engages; remove the covariance term and it slides back toward 1, and the metric would catch it.

### 5. An evaluation designed to falsify the model

- **Generalisation probes**, each varying one factor away from training, against a JEPA in-distribution baseline of 0.087. Held-out high susceptibility raises the error to 0.158, a factor of 1.8, and is the genuine gap: the fast-progressor tail is where the model struggles. A rollout extended to 96 months raises it to 0.126. Genuinely late treatment timing, with the drug starting between months 26 and 40, which training never sees, shows no degradation, because the treatment lever generalises over its start time. (An earlier version of this probe was accidentally in-distribution; I fixed it.)
- **Cross-mechanism transfer.** I added a second valid generator that keeps the state space, contexts, bounds, and treatment and ERCP semantics but changes the coupling equations, and never train on it. It's a falsification probe for generator inversion, not a leaderboard. JEPA degrades from 0.082 to 0.096 and supervised history from 0.072 to 0.085, both about a sixth worse. The models keep their guarantees but do not transfer accuracy cleanly, so the in-distribution result must be read as recovery of one mechanism, not mechanism-independent understanding of the disease.
- **The phase boundary, the central experiment.** As clinic visits thin from monthly to yearly, the history model's advantage over the memoryless model grows monotonically (from about 0.001 to 0.003) but stays a whisper, because  $x(t)$  already carries the susceptibility. Under sensor noise, by contrast, the same history latent denoises the current state to 0.015–0.023 while the raw observation degrades to 0.039–0.114, a decisive win. In fairness this is a win for history, not for the JEPA objective: the plain supervised denoiser (0.023 at noise level 0.15) beats the JEPA one (0.038). A window buys filtering; the latent-prediction objective is not what delivers it.
- **Decompensation detection**, defined as portal hypertension crossing 0.50. Under realistic six-month follow-up, recall is about 0.6 (six or seven of eleven true events), with no false alarms and a timing error of three to four months; the models fall

within one patient of each other, so eleven events are too few to rank. Under a pure 36-month rollout with no re-observation, recall is zero of eleven: the model under-predicts the accelerating portal-hypertension tail, projecting a final value near 0.25 against a true 0.63. The second number is the honest ceiling.

- **A counterfactual**, moving treatment six months earlier, validated against a generator re-run that shares the same noise stream. The true change in malignancy is  $-0.014$  and the model's  $-0.012$ , with sign agreement 0.92 but a per-patient correlation of only 0.51: right direction and rough magnitude, with substantial scatter.

**What this can and cannot establish.** Training is on a single generator, so doing well there means the model recovered that generator's rules. That is the best outcome available, since here the generator is the disease. The alternate mechanism is a controlled structural shift, not an independent clinical source, so it shows the risk rather than resolving it, and none of this speaks to a real liver. A genuine claim of mechanism robustness would need a family of independent generators and, ultimately, external data.

## 6. Why the model predicted decompensation at about month 33

The assignment's worked example asks why the model would call decompensation around month 30; the closest held-out case I have lands a little later. Consider a patient in disease class two, a non-responder with hidden susceptibility 1.91, whose true decompensation occurs in month 33 and whom the model flags at month 36 under six-month follow-up. The prediction is auditable at three levels, all produced by `python -m lwm.explain`. First, an increment ledger: because portal hypertension is the previous value plus a non-negative increment, its entire rise is a printable sum of monthly steps. Second, input-gradient saliency: the prediction leans on the structural fields (portal-hypertension history, strictures, fibrosis, ductopenia) rather than the noisy fast channels. Third, a latent read-back: a linear read of susceptibility from the latent returns 1.87 against the true 1.91, so the model inferred a fast progressor from the twelve-month history, and that drives the steep projection. The honest caveat belongs in the same breath: the call depends on periodic re-observation, and without it the model under-predicts portal hypertension and misses the event. The accurate statement is that, given follow-up data, the model projected portal hypertension across the threshold three months late, not that it foresaw decompensation from month twelve.

## 7. Residual risk and what I would do next

- **The accelerating portal-hypertension tail is improved, not solved.** A calibrated eightieth-percentile supervised-history risk head (pinball loss, conformal offset, coverage 0.73 against a nominal 0.80, so it still under-covers) catches every decompensator under six-month follow-up, eleven of eleven across three seeds, at roughly three false alarms. Under a pure 36-month rollout, though, its recall is  $0.21 \pm 0.17$ , so that tail remains largely unsolved. The same quantile model is also the best point predictor, at  $0.069 \pm 0.005$ .
- **The JEPA objective is unproven, and I tested its best case.** I ran the regime it should most favour, a full forecast under sensor noise and irregular observation together, and the models tie. The denoising advantage that the history latent shows on isolated state estimation does not carry over to multi-step forecast accuracy, an honest negative result that closes rather than opens the door on "JEPA wins under degraded observation".
- **Some evidence is thin, and the three-seed spread is the guard.** The accuracy tie, the denoising advantage, and the cross-mechanism degradation all hold across the three seeds, but the manifold ranking rests on a single critic seed, the decompensation ranking on eleven events with wide confidence intervals, and the counterfactual correlation is only 0.51. The cross-mechanism probe is still a single designed shift; the next step is a family of independently parameterised generators, each with its own interval.
- **Why the headline might be wrong.** The claim that history buys robustness while the JEPA objective buys little rests on a denoising comparison in which the history models were noise-augmented and the memoryless baseline was not given a denoiser. That is a fair structural comparison, because a single state genuinely cannot filter, but it assumes the gap is structural rather than a matter of an under-trained baseline.